

Review Article

CT derived fractional flow reserve: Part 1 – Comprehensive review of methodologies

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ABSTRACT

Advancements in cardiac computed tomography angiography (CCTA) have enabled the extraction of physiological data from an anatomy-based imaging modality. This review outlines the key methodologies for deriving fractional flow reserve (FFR) from CCTA, with a focus on two primary methods: 1) computational fluid dynamics-based FFR (CT-FFR) and 2) plaque-derived ischemia assessment using artificial intelligence and quantitative plaque metrics. These techniques have expanded the role of CCTA beyond anatomical assessment, allowing for concurrent evaluation of coronary physiology without the need for invasive testing. This review provides an overview of the principles, workflows, and limitations of each technique and aims to inform on the current state and future direction of non-invasive coronary physiology assessment.

Abbreviations: CAD, Coronary artery disease; CCTA, Cardiac computed tomography angiography; CT-FFR, Computed tomography-derived fractional flow reserve; FFR, Fractional flow reserve; AUC, Area under the curve; SPECT, Single photon emission computed tomography; PET, Positron emission tomography; CFD, Computational fluid dynamics; CFR, Coronary flow reserve; ICA, Invasive coronary angiography; MACE, Major adverse cardiovascular outcomes; MI, Myocardial infarction; AI, Artificial Intelligence; CVP, Central venous pressure; CPR, Curved planar reformation.

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1. Introduction

Cardiac computed tomography angiography (CCTA) is a well-established non-invasive tool for assessing coronary anatomy, offering high sensitivity and excellent negative predictive value for excluding obstructive coronary artery disease (CAD).¹ However, the specificity of CCTA for identifying functionally significant lesions remains modest.² A substantial proportion of stenoses deemed anatomically severe on CCTA are not hemodynamically significant when assessed with invasive fractional flow reserve (FFR),³ a limitation not unique to CCTA but shared across anatomic imaging modalities, including invasive coronary angiography and intravascular ultrasound.^{4,5}

This disconnect between anatomical and functional severity has driven the development of techniques that extract physiological information from CCTA datasets. The most widely adopted of these is computed tomography-derived fractional flow reserve (CT-FFR), which applies computational fluid dynamics (CFD) to model coronary flow and pressure.^{3,6} More recently, artificial intelligence (AI)-based methods have emerged that integrate plaque morphology and vessel geometry to predict lesion-specific ischemia, often with reduced computational complexity.^{7,8}

This first paper in a two-part series by the FiRST SCCT Committee provides a detailed overview of the methodologies used to derive physiological data from CCTA, including CT-FFR, AI-driven plaque-based ischemia assessment, and other emerging techniques. Part II will focus on these tools' diagnostic performance, prognostic value, and clinical application. In this manuscript, we examine the core principles, technical assumptions, workflows, and limitations that underpin the use of CCTA for non-invasive functional assessment.

1.1. Invasive physiologic assessment of coronary artery disease

The foundation of invasive coronary physiology was established in the 1970s when Dr Gould introduced the concept of coronary flow reserve (CFR)⁹ defined as the ratio of hyperemic coronary flow to resting coronary flow, demonstrating its relationship with varying degrees of coronary stenosis. This seminal work highlighted the limitations of anatomical assessment alone and laid the foundation for functional lesion evaluation.

Following Gould's foundational work, coronary flow reserve was applied clinically using Doppler and thermodilution-based flow wires, though these techniques were technically challenging and less reproducible, which ultimately led to the development of pressure-derived indices such as FFR.

Fractional flow reserve uses intracoronary pressure measurements to quantify the hemodynamic significance of coronary stenoses.¹⁰ FFR was initially defined as the ratio of maximal coronary flow in the presence of stenosis to the hypothetical flow in its absence. Since direct flow measurement is impractical in most settings, pressure-based surrogates were adopted. Under maximal hyperemia conditions where microvascular resistance is minimal and constant, the flow becomes directly proportional to pressure. Thus, FFR is clinically approximated as the ratio of distal coronary pressure (Pd) to aortic pressure (Pa):

$$FFR = P_d / P_a$$

FFR provides a lesion-specific index of flow limitation under hyperemic conditions. For example, an FFR of 0.65 indicates that maximal coronary blood flow is reduced to 65 % of what it would be in the absence of any stenosis.^{11,12}

It is essential to recognize that coronary physiology is influenced not only by epicardial stenosis but also by the integrity of the coronary microvasculature. During maximal hyperemia, microvascular resistance is assumed to be minimal and uniform, a key assumption of FFR measurements.^{13,14} However, this may not hold in conditions associated with microvascular dysfunction, such as diabetes, longstanding hypertension, left ventricular hypertrophy, aortic stenosis, and

post-myocardial infarction microvascular stunning.¹⁵ These conditions can blunt the hyperemic response, potentially leading to underestimation of ischemia. Reduced diagnostic accuracy is particularly relevant when FFR values fall near the clinical threshold of 0.80, where slight variations in FFR may influence management decisions. This reinforces the importance of interpreting FFR as a continuous risk measure rather than a binary test, especially in borderline cases. Nonetheless, in appropriately selected patients, FFR remains a robust, lesion-specific tool for assessing the functional significance of epicardial stenoses and is widely accepted as the invasive reference standard.

FFR has become the invasive reference standard for assessing the hemodynamically significant coronary lesions. Multiple clinical trials have demonstrated that FFR-guided revascularization reduces unnecessary percutaneous coronary interventions and improves patient outcomes.¹⁶ Coronary stenosis with an FFR ≤ 0.80 is generally considered hemodynamically significant and, therefore, may be managed with revascularization.¹⁷ However, more recent trials, such as ISCHEMIA, RIPCORD-2, FUTURE, and FAME-3; have challenged the assumption that all FFR-positive lesions benefit from revascularization.^{18–21} These studies suggest that revascularization should be considered selectively, particularly in patients with persistent angina despite optimal medical therapy, rather than applied uniformly based on FFR values alone. In this context, emerging data on the physiological pattern of disease provide additional nuance. The Pullback Pressure Gradient (PPG), quantifies the pattern of coronary artery disease based on FFR pullbacks, distinguishing between focal and diffuse disease.²² Studies have shown that FFR-positive lesions with focal disease (high PPG) are more likely to benefit from PCI, whereas diffuse disease (low PPG) is associated with less symptomatic and physiological improvement after revascularization.^{23,24} Incorporating lesion-level anatomical and physiological patterns alongside FFR may improve patient selection and help reconcile conflicting evidence across recent trials.

1.2. Non-invasive physiologic assessment of coronary artery disease

Recent advancements in CT scanner technology, image reconstruction and machine learning have significantly improved the ability to derive non-invasive physiological information from CCTA. The improvement spatial and temporal resolution coupled with AI tools to automated coronary segmentation have permitted more accurate and efficient identification of coronary luminal geometry, plaque volume and characteristics.

The ability to apply CFD to routine CCTA scans has transformed CCTA from a purely anatomical tool into one capable of providing non-invasive hemodynamic assessment via CT-FFR. In parallel, emerging methods that leverage AI and plaque morphology offer alternative pathways to estimate ischemia without flow modelling.

The following sections describe principal methodologies currently deriving physiology from CCTA.

1. CFD derived FFR from CCTA.
2. Plaque-based ischemia and FFR.
3. Deep Learning Algorithms Supporting FFR from CCTA.

1. CFD derived FFR from CCTA

Technological advances in CFD and 3-dimensional (3D) image-based modelling have enabled the non-invasive measurement of FFR from pre-existing CCTA data without additional imaging or medications.^{25,26} CFD applies numerical analysis to solve equations governing fluid flow, such as velocity, density, and pressure and has a wide range of applications, including coronary flow assessment. Evaluation of coronary flow and determination of CT-FFR using these mathematical models requires four fundamental steps (Fig. 1).

- Coronary Segmentation
- Myocardial Mass Calculation

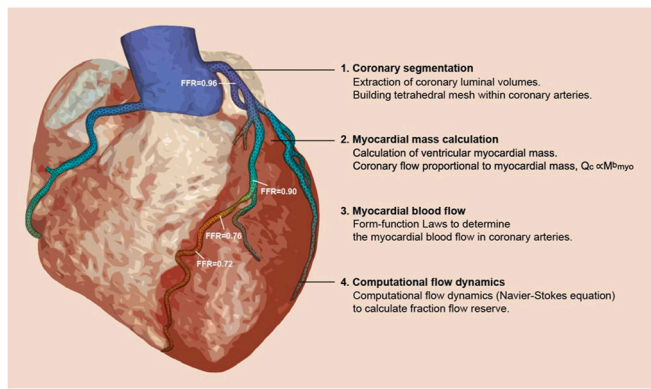


Fig. 1. Pictorial representation of four fundamental steps to derived CT-FFR using computational fluid dynamics methodology.

- Myocardial blood flow estimation
- Computational fluid dynamics simulation

Step 1: Coronary Segmentation.

Contemporary image acquisition techniques and spatial and temporal resolution have enabled the reconstruction of reliable three-dimensional (3D) anatomical models of the coronary arteries using ≥ 64 -detector row CCTA. Image reconstruction using a 512×512 matrix achieves a spatial resolution of approximately 0.5 mm.²⁷ The reconstruction of luminal volumes of the coronary arteries is essential in assessing CT-FFR.

Anatomy plays a central role in CT-FFR assessment, given the universal “form-function” relationship of the circulatory system, wherein anatomical structure reflects the physiological function. Flow in the coronary arteries is contingent upon lumen size. Thus, the total per-patient coronary luminal volume facilitates determining total coronary flow, total resistance, relative flow, and resistance in individual coronary arteries. CFD requires a discrete finite set of 3D volumes within the coronary dataset to implement the laws of fluid dynamics. Consequently, millions of tetrahedral “mesh” elements are generated within the coronary artery tree (using the finite element method), each of which is solved for velocity and pressure using powerful computing workstations.²⁸

Theoretically, accurate pressure and flow would be obtained if the governing laws of fluid dynamics were applied to the entire circulatory system (the heart and more than 5 billion blood vessels in the human circulation). However, as this is impractical, the anatomical domain of interest is restricted to the coronary branches and aortic root. Moreover, the various components of the circulatory system must be defined by appropriate boundary conditions. Three types of boundary conditions are identified for the assessment of CT-FFR of the coronary arteries.

- i) Inlet Boundary: Represents blood entering the aorta.
- ii) Aortic Outlet: Represents the mean pressure within the aorta.
- iii) Coronary Outlets: Represent the resistance at each coronary branch.

The lumped-parameter models are subsequently integrated with the aortic root and coronary arteries that have been extracted from the CCTA to compute these values. These models simulate the interaction between the heart and the systemic circulation, facilitating the calculation of patient-specific cardiac output and mean aortic pressure, which are further refined using allometric scaling principles.^{25,28,29}

Step 2: Myocardial Mass Calculation.

Allometric scaling laws describe the biological scaling relationships, that is, physical quantities that scale with each other over a substantial interval. In the context of coronary physiology, these laws explain how an object's size (or mass) is related to its anatomical and physiological

characteristics, for example, the relationship between myocardial mass and coronary blood flow or coronary volume.^{25,26}

Biological allometric scaling laws are mathematically represented as follows:

$$Y = Y_0 M^b$$

where: M: mass, b: scaling exponent, and Y_0 : normalization constant.

These scaling laws are integral to the subsequent steps, as they provide a foundational framework for quantifying total and vessel-specific coronary flow and resistance. Under resting conditions, total coronary flow is directly proportional to myocardial mass, expressed as $Q_c M^b_{myo}$ (Q_c : coronary flow, M_{myo} : myocardial mass, b: scaling exponent). In this equation, myocardial mass is an important variable that can be accurately calculated from CCTA data, providing input for the subsequent physiological assessments.²⁵ Applying these scaling laws, the myocardial mass derived from CCTA estimates the total coronary flow and resistance, which are fundamental in assessing coronary artery disease's physiological significance.

Step 3: Myocardial Blood Flow Calculation.

Once the myocardial mass has been determined, total coronary flow can be quantified by incorporating this information with an assumed mean aortic pressure. This allows for calculating total basal coronary resistance, a required variable in assessing coronary physiology.²⁵

(i) Vessel specific resistance

Resistance at each coronary outlet (branches) is calculated using the morphometric principles, which reflect the intrinsic form-and-function relationship of the coronary vasculature. The flow through each branch of the epicardial coronary artery is modelled as proportional to its luminal area, as determined by the CCTA. Thus, using the total basal coronary resistance, the resistance at each coronary outlet is estimated based on the assumption that resistance at rest is inversely proportional to the coronary diameter.²⁹

(ii) Computation of hyperemic coronary resistance

To simulate maximal hyperemia, microvascular resistance is scaled to approximately 24 % of its resting value. This estimate is based on data from intravenous adenosine infusion at 140 $\mu\text{g/kg/min}$, as reported in invasive studies.³⁰ This value is used in the boundary conditions to simulate hyperemic flow.³¹

Step 4: Computational fluid dynamics.

The physical principles that govern blood flow in the coronary arteries are described by the Navier–Stokes equations, which express the conservation of mass and momentum, which is essentially a fluid dynamics application of Newton's second law ($F = ma$). In large vessels, blood behaves similarly to a Newtonian fluid with constant viscosity, allowing these equations to be applied. While the underlying physics is conceptually straightforward, solving the Navier–Stokes equations in practice is highly complex. It requires the coronary anatomy to be segmented and broken into millions of discrete mesh elements, across which the equations are numerically solved to compute pressure and flow. This computational approach is the foundation of CFD-based CT-FFR and demands substantial processing power to solve millions of interdependent equations across the coronary tree.²⁵

The integration of anatomy, physiology, and CFD facilitates the computation of CT-FFR (Figs. 1 and 2). After creating the vascular domain (anatomical model) and defining the boundary conditions (flow and resistance) at rest and hyperemia, CFD can compute the hyperemic pressure and flow across the vascular domain and during the cardiac cycle. From these values, the mean hyperemic pressure is calculated. CT-FFR is derived by normalizing the mean hyperemic pressure field at a particular point by the average mean hyperemic pressure in the aorta (i.e., $\text{CT-FFR} = P_d/P_a$). The result enables the evaluation of CT-FFR

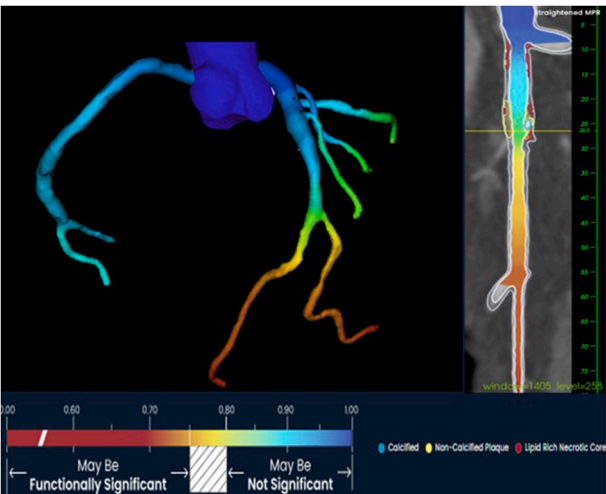


Fig. 2. Plaque derived FFR along with vessel in straightened curved planar reformation (CPR) view showing plaque distribution and characteristics contributing to the significant drop in FFR values. Image generated by (vascuCAP®, Elucid Bioimaging, Inc., Boston, MA).

across stenosis and at every point along the length of the coronary vessel, thereby allowing the generation of a colour-coded map of CT-FFR along the entire coronary artery. Currently, the most established clinically available method is FFR_{CT} (HeartFlow, Redwood City, USA), although other less-established techniques are also available or in development. There are several limitations of this technique, which are summarized in Table 1.

Using computational fluid dynamics (CFD) to derive CT-FFR values also enables the extraction of other hemodynamic indices related to the coronary geometry. CT-derived metrics such as wall shear stress (WSS) and axial plaque stress have been previously derived from a combination of anatomical features and CFD-derived hemodynamics. The study by Lee, Joo et al.³² highlighted that lesions with adverse plaque morphology and high-risk hemodynamic characteristics, including elevated WSS and axial plaque stress, were more likely to be associated with plaque rupture. Notably, the change in CT-FFR (Δ CT-FFR) across a lesion was shown to be a strong predictor of plaque rupture, suggesting that pressure loss may serve as a surrogate for more complex hemodynamic parameters such as WSS. This is particularly relevant in clinical practice, as

Table 1
Limitations of CFD based CT-FFR techniques.

Assumption	CFD-Based CT-FFR Limitation
Coronary artery is a Rigid Tube	Ignores arterial compliance and deformation due to pulsatile blood flow.
Blood pressure is Same for Everyone	Does not account for individual variations in systemic and coronary pressures.
Microvascular resistance is Same for Everyone	Ignores patient-specific microvascular physiology and variability.
Hyperemic flow is the Same for Everyone	Overlooks individual differences in coronary flow reserve.
Microvascular resistance is 0 mm Hg at hyperemia	An unrealistic assumption, as resistance does not drop to zero.
Pressure drop across a stenosis is linear	Oversimplifies complex hemodynamics, leading to potential inaccuracies.
No collateral circulation Exists	Ignores alternative perfusion pathways, which can significantly impact ischemia assessment.
All flow is laminar	Neglects turbulent and disturbed flow patterns that occur in diseased arteries.
Blood is Newtonian (no Rheology)	Ignores the non-Newtonian behavior of blood, affecting shear stress calculations.
What occurs in epicardial artery applies to myocardium	Does not consider microvascular dysfunction and heterogeneous myocardial perfusion.
Coronary calcification	May affect the precise boundary detection which is essential for CFD based CT-FFR

Δ CT-FFR can be readily measured from the 3D CT-FFR model without additional complex computations. Integrating these hemodynamic markers into routine assessments could improve risk stratification and early identification of high-risk plaques; however, requires further validation in prospective studies.

The accuracy of CT-FFR computation is highly dependent on the precision of the 3D anatomical model of the coronary geometry. Advances in CT technology, including high-resolution photon-counting detector scanners, motion correction algorithms, and super-resolution deep learning reconstruction, can help address some of these challenges. These innovations enable better delineation of coronary anatomy and may improve CT-FFR accuracy. Future studies are needed to further evaluate these technologies.

The clinical evidence supporting the use of CT-FFR is the focus of Part II of this review series.

2) Plaque-Based Ischemia Assessment from CCTA

Coronary CT angiography has evolved from a purely anatomical imaging tool to one capable of characterizing plaque morphology and composition in detail. Beyond stenosis severity, features such as low-attenuation plaque, positive remodelling, and necrotic core volume are now recognized as important predictors of lesion-specific ischemia and adverse outcomes.^{33,34}

Recent advances in artificial intelligence and machine learning have enabled non-CFD-based approaches to estimate the likelihood of ischemia directly from plaque characteristics extracted from CCTA.^{7,8} These data-driven models bypass the need for complex flow simulation and instead infer functional significance based on morphological and compositional features of atherosclerotic plaque. While they do not simulate coronary hemodynamics, they aim to provide fast, physiology-based outputs from a standard CCTA scan.

Two primary methodological approaches have emerged.

The first is a non-CFD-based CT-FFR technique that estimates lesion-specific ischemia by analyzing vessel geometry, plaque burden, and compositional features derived from CCTA (vascuCAP, Elucid Bioimaging) (Fig. 2).³⁵ This method applies machine learning algorithms trained on plaque characterization data derived from imaging-histology correlations. The model incorporates a range of variables, including lesion length, total plaque volume, minimal lumen area, and the relative proportions of fibrous, lipid-rich, and necrotic core tissue.³⁵ These features are integrated to infer the likelihood of impaired flow across lesions under hyperaemic conditions. The output is provided as a continuous value resembling CT-FFR without simulating coronary flow or applying CFD. The physiological inference is based on the premise that certain plaque morphologies and compositions are more likely to compromise vasodilatory capacity and flow reserve, which has been supported by correlations with invasive FFR in validation studies.³⁵

The second approach represents a classification-based model of ischemia prediction that uses AI to categorize lesions as ischemic or non-ischemic based on plaque and vessel characteristics (AI-QCT Ischemia, Cleerly) (Fig. 3a,b).⁷ This methodology evaluates over 30 quantitative features such as plaque burden, lesion length, low-attenuation plaque, and minimal lumen area (6). These algorithms generate a binary (yes/no) output for each lesion based on statistical models trained against invasive FFR datasets. While this format may facilitate faster decision-making, it does not provide a continuous measure of flow limitation like traditional CT-FFR techniques or support detailed hemodynamic interpretation.

A related classification-based model developed by Lin et al. similarly integrates multiple quantitative plaque features to predict lesion-specific ischemia from CCTA.³⁶ Using 19 inputs—including low-density non-calcified plaque, diameter stenosis, and contrast density difference—the model achieved diagnostic performance comparable to CT-FFR and superior to visual stenosis in external validation against invasive FFR and PET-derived myocardial blood flow.³⁶ While

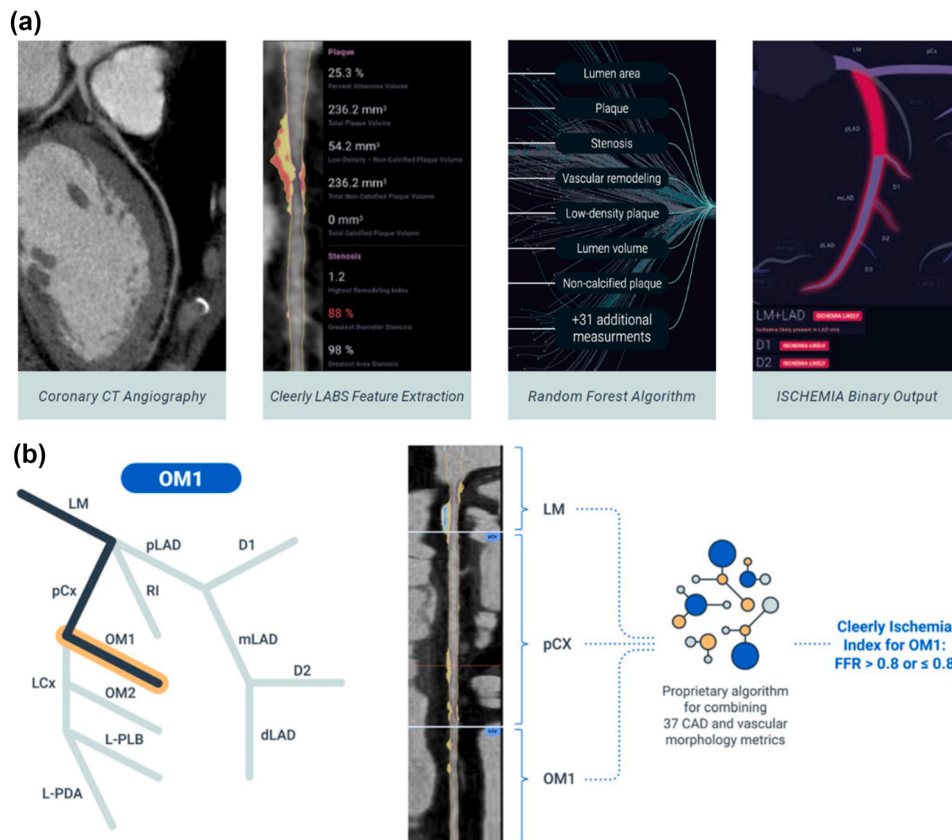


Fig. 3a. ISCHEMIA uses numerous quantitative measures of atherosclerosis, stenosis, and vascular morphology derived from Cleerly LABS analysis of a CCTA as inputs to its AI algorithm. The algorithm generates a binary determination of likely ischemia (estimated invasive FFR < 0.80) or unlikely ischemia (estimated Invasive FFR > 0.80) in a vessel specific fashion, and is displayed along with other actionable measures of atherosclerosis and stenosis on vessel/lesion specific maps and report. Image generated by (Cleerly Health, Inc).

Fig. 3b: ISCHEMIA prediction for a 1st obtuse marginal (OM1) branch of the left circumflex artery will compromise the subtended assessment of the left main (LM), proximal left circumflex (pCx) and OM1 branch vessel. All quantitative measures of atherosclerosis, stenosis, and vascular morphology of each coronary segment leading to and including the analyzed vessel are included as inputs to the algorithm.

the model produces a continuous machine-learning score, this output is not equivalent to FFR and cannot be directly translated to physiological thresholds used in clinical decision-making. Instead, a diagnostic cut-off is applied to yield a binary classification of ischemia. Notably, the model incorporates explainable AI techniques to quantify the relative contribution of each feature to a given prediction.³⁶ This may improve clinician interpretability and better support integrating AI-based ischemia models into clinical workflows.

These plaque-based methods offer several advantages. They are computationally efficient, require shorter processing times than CFD-based methods, and are not subject to assumptions inherent in fluid dynamics modelling, such as predefined boundary conditions, simplified blood viscosity, or uniform microvascular resistance.^{7,35} By incorporating plaque morphology, they reflect additional biological processes, such as endothelial dysfunction and altered vasodilatory capacity, which may not be captured by stenosis severity alone.

However, important limitations remain. These approaches are based on static anatomical features and do not model dynamic flow or pressure responses. Binary classification tools, while efficient, may limit the ability to localize lesion-level flow limitation or assess disease severity along a continuum. Moreover, validation of these methods across diverse clinical populations and complex coronary anatomies is still evolving, and they currently lack the volume and breadth of prospective trial data available for CFD-based techniques.^{37–39}

By contrast, CFD-based CT-FFR provides patient-specific, lesion-level simulation of coronary haemodynamics. It yields continuous FFR values, allows assessment of tandem and diffuse disease, and supports procedural planning through virtual stenting simulations. CT-FFR has been validated against invasive FFR across multiple prospective multicentre trials, establishing its diagnostic accuracy and prognostic value in a wide range of clinical settings.^{3,25}

Nonetheless, plaque-based ischemia assessment represents a promising and complementary direction. As these algorithms continue to evolve and are tested in larger studies, they may enhance the utility of CCTA as a single, comprehensive tool for anatomical and functional assessment of coronary artery disease.

3. Deep Learning Algorithms Deriving FFR from CCTA

Recent advances in artificial intelligence have introduced an alternative to traditional CFD-based CT-FFR, using deep learning algorithms to estimate lesion-specific FFR directly from standard CCTA images. These models are typically trained using large datasets that pair coronary anatomy with FFR values, derived either from CFD simulations or invasive measurements. Rather than solving the governing fluid dynamic equations, these data-driven models aim to infer physiological significance based on learnt patterns between anatomy and haemodynamics.⁴⁰

One such example is DeepVessel FFR (Keya Medical), which uses a multi-stage deep learning pipeline to process segmented coronary anatomy and output predicted FFR values along the vessel tree. The algorithm has been trained using CFD-derived FFR data validated against invasive measurements, allowing it to approximate physiological gradients without modelling flow explicitly. DeepVessel FFR can be deployed as an on-site software tool, offering relatively rapid turnaround (typically within 5 min). While promising, the software still requires human input to verify coronary segmentation and approve final results before clinical use.

Early studies have reported high diagnostic accuracy compared with invasive FFR (e.g., AUC 0.93 in a single-centre study of 93 patients), but these findings should be interpreted cautiously given the small sample size and lack of prospective multi-centre validation.⁴¹ The recently

published TARGET trial assessed the clinical utility of this approach in real-time and reported a reduction in unnecessary invasive angiography and improved alignment between anatomical and functional disease.⁴²

The diagnostic performance of this and other deep learning-based CT-FFR has been summarized in a recent meta-analysis of 17 studies encompassing 3255 patients. The pooled per-patient sensitivity was 86 % and specificity was 87 % when compared to invasive FFR as the reference standard.⁴³ The authors highlighted ongoing concerns regarding sample diversity, standardisation across imaging platforms, and lack of large-scale prospective validation. Most current datasets are derived from East Asian populations and may not adequately represent global variations in scanner types, acquisition protocols, or patient comorbidities. Furthermore, segmentation algorithms used in these models may be affected by calcification, motion artefacts, or suboptimal contrast enhancement, limiting their generalisability in real-world clinical settings.

Overall, deep learning-based CT-FFR offers a scalable, on-site alternative to traditional CFD techniques. While the initial diagnostic performance is encouraging, further validation across broader patient populations and anatomical subsets is needed.

2. Conclusion

The expanding role of CCTA now encompasses not only anatomical delineation but also non-invasive assessment of coronary physiology. CT-FFR enables lesion-level quantification of flow limitation with strong validation across multiple prospective trials, while plaque-based models leverage artificial intelligence to integrate anatomical and compositional plaque features, offering computational efficiency and broader scalability. Each technique presents unique strengths and limitations: CFD-based approaches provide detailed hemodynamic assessment but are resource-intensive and depend on modelling assumptions, whereas plaque-based methods incorporate biologically relevant features of atherosclerosis but currently lack equivalent validation and may be less precise in localizing flow-limiting lesions. As imaging technology and machine learning continue to evolve, hybrid frameworks that combine anatomical, morphological, and physiological information from a single CCTA dataset may improve diagnostic accuracy, enhance risk stratification, and support more tailored management of coronary artery disease.

Author disclosures

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Declaration of competing interest

The authors declare that there are no conflicts of interest.

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